Statistical Business Analysis Report

Objective: Analyze business sales data using descriptive statistics, hypothesis testing, correlation, confidence intervals and regression.

1. Dataset Overview

Metric	Value
Number of records	100
Number of columns	7
Average sales	123,650.48
Median sales	97,955.50
Sales standard deviation	100,161.09
Minimum sales	6,540.00
Maximum sales	373,932.00

2. Confidence Interval

95% confidence interval for mean sales: 103,776.35 to 143,524.61.

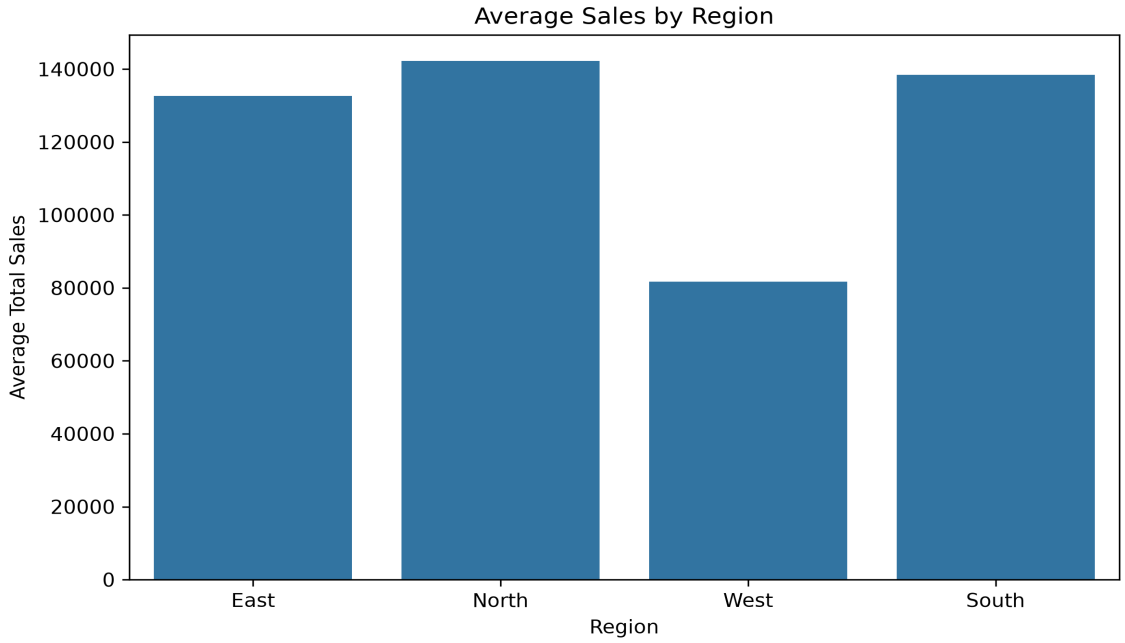
3. Correlation Analysis

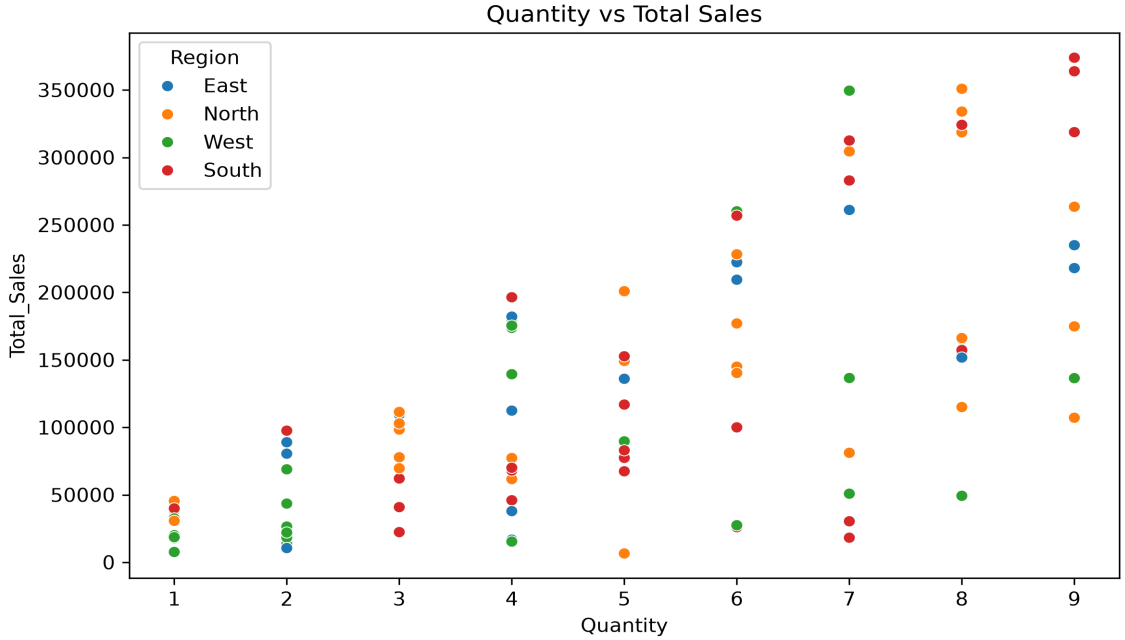
Correlation between quantity and total sales: 0.6881.

4. Regression Analysis

R-squared: 0.8839. The model estimates the association of quantity and price with total sales. Interpret results in light of the dataset's sales calculation.

5. Sales Visualizations





6. Business Recommendations

Compare average sales across regions and product categories. Use the hypothesis-test results to identify statistically supported differences. Investigate high-performing products and regions before making inventory or marketing decisions. Statistical association alone does not prove causation.